Safe Fast Charging Robust to Battery Aging: A Solver-Free Safety Filter Without a Health Oracle

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Abstract*—Lithium-ion fast charging must hold hard limits on cell temperature (below 45 °C) and voltage (below 4.2 V) at every instant; past them a cell risks thermal runaway and lithium plating. Learning-based chargers encode these limits as a soft penalty the policy trades away, their safety decays as the cell ages, and the safety layers that avoid this need an online solver and an identified state-of-health (SOH). We present a real-time safety filter that removes both needs. Zero current arrests heating and drops voltage to open circuit, so it is a universal safe backup that collapses the predictive safety filter to a one-dimensional projection a bisection solves on a reduced-order model (ROM), with no QP. Safety is monotone in the two aging channels, resistance growth and cooling loss, so a conservative datasheet bound certifies it and no parameter is identified; an online estimator relaxes the bound to recover charge, and the guarantee never depends on the estimate. Plating has no universal backup, so we enforce it with a temperature-dependent cap and report it as enforced, not proven. We validate against a Doyle–Fuller–Newman plant in simulation, replaying the controller's realized current so the ROM never self-reports safety (voltage RMSE 20.3 mV). A trivial greedy governor matches a trained reinforcement-learning policy on every metric, so the filter, not the learner, carries safety. The filter records no observed temperature or voltage violation where fast CC–CV overshoots 45 °C in every episode, peaking at 40.4 °C, delivers the charge of CC–CV at 1C (SOC 0.773) while removing that protocol's 22% plating, and holds temperature and voltage safe on 156/156 disjoint aged draws (Clopper–Pearson 1.9%) and under a full recalibration of the ROM to a second cell chemistry (100/100, voltage RMSE 26.5 mV). Each projection costs at most 20 model evaluations and 904 B, with no solver.

***Index Terms*—Fast charging, safety filter, control-invariant set.**

NOMENCLATURE

T, V	Cell temperature and terminal voltage; limits $T_{\max}{=}45\,C$, $V_{\max}{=}4.2\,V$.
$\phi_{\rm an}$	Anode plating overpotential, $\phi_s{-}\phi_e$ at the separator.
I	Applied current; I_π proposed, $I_{\rm cmd}$ commanded, $I^\star$ largest admissible.
$\mathcal{S}$	Certified safe set $\{T \le T_{\max}, V \le V_{\max}\}$; $\mathcal{S}_{-\delta}$ its margin-tightened inner approximation.
g_c	One-step peak of constraint c predicted by the ROM.
δ_T, δ_V, δ_P	Control margins on temperature, voltage, plating; δ_{T0} the base thermal margin.
s_R, γ	Aging channels: resistance scale and cooling-loss factor; $\hat{s}_R$ estimated, $s_R^\star$ the rated end-of-life bound.
K, f	Margin gains: resistance throttle and cooling reserve.
Δt, $\tau_{\rm RC}$	Control step and RC time constant, both 30 s.
$Q_{\rm gen}$	Heat generation; ohmic, activation and reversible parts $Q_{\rm ohm}$, $Q_{\rm act}$, $Q_{\rm rev}$.
$C_{\rm th}$, hA	Lumped thermal capacity and heat-transfer coefficient.
R_Ω, R_1, V_1	Ohmic resistance, RC-branch resistance and its overpotential.
ϵ, n	Bisection current resolution; episodes in a stress test.

I. INTRODUCTION

A fifteen-minute electric-vehicle charge draws current at rates that push a lithium-ion cell toward two irreversible failures: thermal runaway and lithium plating on the graphite anode [1], [2]. Both are governed by hard limits: temperature below $45\,C$, terminal voltage below $4.2\,V$, and a non-negative anode overpotential. A fast charger is therefore a controller that must respect these limits as *hard* constraints, not soft penalties, the prerequisite for deployment in a battery management system (BMS).

The industrial default, constant-current–constant-voltage (CC–CV), is a fixed rule that is safe only when tuned slow. Deep reinforcement learning (RL) promises the missing adaptivity, and many recent chargers apply it [3], [4], [5]. Three gaps remain. First, most encode safety as a soft penalty the agent trades away for reward; the few with a safety layer lean on a learned surrogate whose guarantee is only statistical. Second, papers rarely state which constraints a filter can certify. Temperature and voltage react instantly to current, whereas plating is history-dependent. Third, safety is checked almost exclusively on fresh cells, and a policy fitted to fresh dynamics turns silently unsafe as internal resistance grows with age.

We close these gaps with a single object, not a new learner. Our contribution is threefold.

- **A solver-free safety filter for temperature and voltage.** Zero current arrests heating and lowers voltage, so it is a globally available backup that reduces the predictive safety filter [6] to a one-dimensional control-invariant projection a bisection solves, with no online optimizer or QP (Prop. 1). The certificate covers temperature and voltage, the two constraints a one-step filter carries; the history-dependent plating limit stays out of it, held in the realized profile by a temperature-current cap, not certified.
- **Aging without a health oracle.** Safety on temperature and voltage is *monotone* in the two aging channels, resistance growth and cooling loss (Prop. 2), an instance of the order-preserving structure of monotone control systems [7], [8]. A conservative *bound* on each channel then certifies safety with no parameter identified: a fail-safe datasheet end-of-life prior supplies the bound, and an online estimator only relaxes it to recover charge. The guarantee never depends on the estimate; we report the certified floor and the estimator's added charge as separate numbers.

- **Cross-chemistry generalization.** Safety holds not only across perturbations of one parameterization but under a *full recalibration of the ROM to a different cell chemistry*: the guarantee is a property of the method, not of one cell.

The filter certifies safety for *any* input it projects, so the guarantee is policy-agnostic (Fig. 1). Sent through it, a trained PID-Lagrangian PPO policy and a trivial greedy governor match on every metric (Sec. VI). A standard adaptive predictive safety filter, by contrast, solves a QP and identifies the aged parameters at every step. Our two mechanisms remove both: the I=0 backup collapses the QP to a scalar bisection (Prop. 1), and monotonicity trades identification for a conservative overbound a fail-safe prior supplies for free (Prop. 2), a cheaper and more robust route to the same safety (Sec. VI-E).

II. Related Work

Safe reinforcement learning enforces constraints through soft penalties [3], Lagrangian relaxation [9], [10], action shielding [11], and learned safety layers [4], [12], each supplying a statistical guarantee fitted to fresh dynamics; charging applications follow the same pattern [13], [14], [5]. **Safety filters** certify or repair a proposed input against a control-invariant set [15]: control barrier functions [16], reference governors [17], Hamilton–Jacobi reachability [18], and the predictive safety filter [6]. We specialize the last to battery charging, where a trivial backup collapses it to a scalar projection. **Charging controllers** span CC–CV, multi-stage CC, model predictive control, and Pontryagin-based optimal profiles [19], [20]; the model-based methods minimize loss directly but need an accurate online model that aging erodes [21], [22]. No prior charging method combines hard T/V limits, an enforced plating constraint [23], [24], resilience to aging [25], freedom from an SOH oracle, and real-time operation.

III. Problem Formulation

We charge a single LG M50 cell (NMC811/graphite-SiOx, 5 Ah), parameterized by the electro-thermal set of Chen et al. [26] and the degradation set of O'Kane et al. (SEI plus partially reversible plating) [27], from 10% to 80% SOC within 40 min from initial temperature T_0, warm ($25\,C$, thermally limited) and cold ($10\,C$, plating limited). The *guaranteed* safe set is

$$\mathcal{S} = \{\, T \le T_{\max},\ V \le V_{\max} \,\}, \quad T_{\max} = 45\,C,\ V_{\max} = 4.2\,V. \tag{1}$$

Here T is the volume-averaged cell temperature both the ROM and the DFN report. Both constraints are *instantaneously stoppable*, since zero current halts heat generation and drops V to the open-circuit voltage, which makes the control-invariance argument below exact for them in the ROM. We also *monitor* the plating overpotential $\phi_{\rm an} = \phi_s - \phi_e$ at the separator interface ($\phi_{\rm an} < 0$ signals plating) [23], but plating is history-dependent and not stoppable, so no one-step margin certifies it. We instead enforce a temperature-dependent current cap, keeping $\mathcal{S}$ certified and plating only empirically enforced.

The state $x = (\mathrm{SOC}, T, V, I_{\rm prev}, t/D, T_{\rm amb})$ uses measurable signals only, the action is the target current $I \in [0, I_{\max}]$ under a slew-rate limit, the reward pays for charge net of wall-energy loss, and the cost is constraint exceedance, a CMDP [28].

IV. Method

A. Solver-free control-invariant projection

From the current state x, let $g(s_R, I)$ be the one-step reduced-order-model (ROM) prediction of $(T, V, \phi_{\rm an})$ under resistance scale s_R and applied current I. The filter projects the policy's proposed current I_π onto the largest admissible value,

$$I_{\rm cmd} = \max\{\, I \in [0, I_\pi] : g(s_R, I) \in \mathcal{S}_{-\delta} \,\}, \tag{2}$$

where $\mathcal{S}_{-\delta}$ tightens $\mathcal{S}$ by control margins δ. Each constraint is monotone in I, which makes the feasible set an interval $[0, I^\star]$ (Fig. 2). Bisection then finds $I^\star$ in $O(\log 1/\epsilon)$ model evaluations, with no matrix solve or rollout.

Proposition 1 (One-step control invariance in the ROM, up to δ_T). *For a control step $\Delta t \gtrsim \tau_{\rm RC}$, the input $I = 0$ makes the input-driven heat vanish and drops V toward the OCV, leaving only a decaying RC residual bounded by $e^{-\Delta t/\tau_{\rm RC}}$ that the margin δ_T absorbs. Thus $I = 0$ is a feasible backup and $\mathcal{S}$ is control-invariant in the ROM* up to δ_T. *That is, if $x_0 \in \mathcal{S}_{-\delta}$ and every step keeps the predicted $(T, V) \in \mathcal{S}_{-\delta}$, then $x_t \in \mathcal{S}$ for all t. The invariance is thus margin-absorbed, not unconditional, and DFN replay verifies the margin end to end.*

Proof. At $I = 0$ every input heat source vanishes and the lone RC residual V_1^2/R_1 decays by $e^{-\Delta t/\tau_{\rm RC}} \approx 0.37$ per step, bounded by δ_T. With Newtonian cooling $\dot{T} \le 0$ up to δ_T whenever $T \ge T_{\rm amb}$ and $V \to \mathrm{OCV} \le V_{\max}$, the input $I = 0$ is a feasible backup at every $x_t \in \mathcal{S}$. The filter admits $I_{\rm cmd}$ only when the one-step prediction lies in $\mathcal{S}_{-\delta} \subseteq \mathcal{S}$, so $x_t \in \mathcal{S}$ for all t by induction.[1] □

Our step ($\Delta t = \tau_{\rm RC} = 30\,s$) meets the condition at one time constant, leaving ~ 0.37 of the RC overpotential energy after the cut in continuous time, which the margin δ_T (base 0.5 °C plus a ∼4.25 °C cooling reserve) absorbs; the discrete update leaves less still, so the bound is conservative. Two opposing requirements pin the step rather than tuning: the residual argument wants $\Delta t \gtrsim \tau_{\rm RC}$, while the explicit discretisation of the RC branch is stable only for $\Delta t \le \tau_{\rm RC}$. Safety is not fragile in the choice: sweeping Δt from 5 s to 120 s yields no violation, moves delivered SOC by under 0.1 points, and never lets the residual exceed 10% of the base margin. The invariance is exact in the *ROM*, and transfer to the DFN plant is bounded by the margins and confirmed by replay (Sec. VI), not asserted, while the plating potential, carrying diffusion memory, stays excluded from $\mathcal{S}$ and enforced empirically.

We keep "certified" scoped to what it proves: exactly one theorem, in the lumped ROM, on volume-averaged temperature, through the high-current phase where $\hat{s}_R \ge s_R^\star$ is pinned. Everything else is simulation-validated with one-sided Clopper–Pearson bounds.

[1] An extended version gives the step-size derivation, the full margin derivation, and figure galleries.

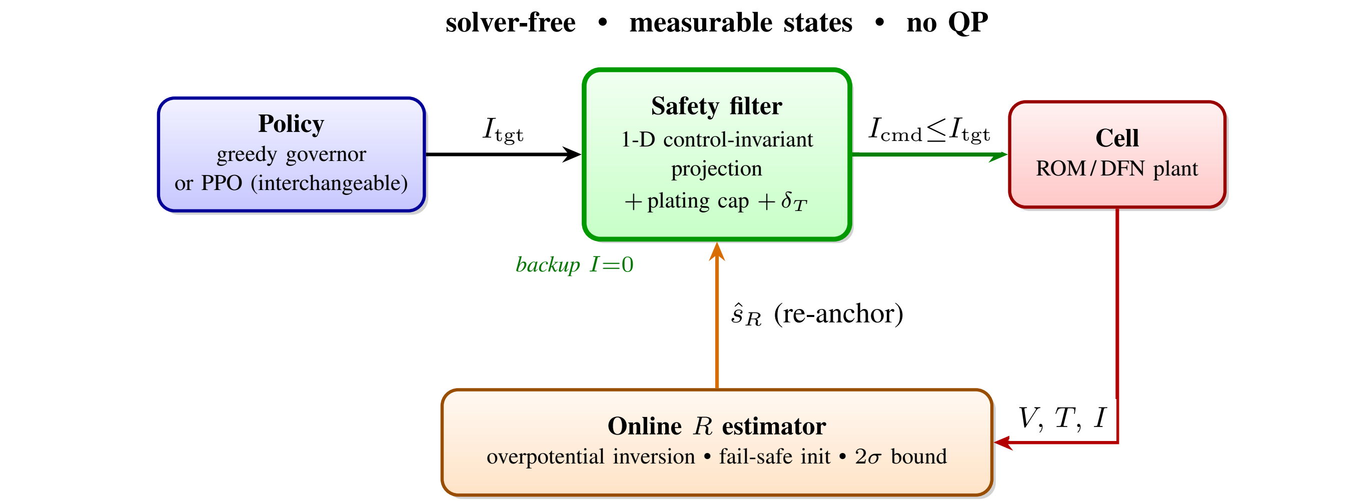

Fig. 1. Control architecture. The policy proposes a target current, the safety filter projects it to the largest value that keeps the one-step transition inside the safe set, and the online estimator re-anchors the filter to the aging cell. Only the filter carries safety.

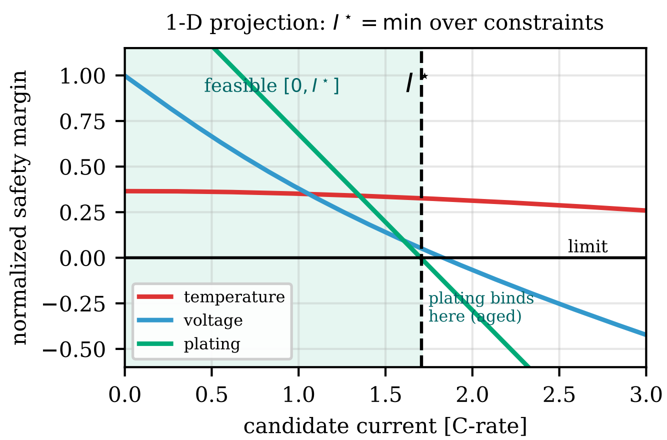

Fig. 2. The solver-free projection. Each hard constraint is monotone in the applied current, so the admissible set is a single interval and the certified cap is $I^\star = \min$ over the per-constraint roots, found by bisection with no QP.

B. Plating as a monotone constraint

The anode overpotential is affine and decreasing in current, so plating enters the same monotone bisection as T and V with a certified cap wherever a nonnegative one exists. The $I{=}0$ backup, however, does not clear the plating margin at the cold, aged, high-SOC corner, so plating is not control-invariant on its own. We keep the trajectory feasible with a temperature-dependent current cap (2 C at $25\,C$ tapering to 1.0 C at $10\,C$ with a 0.70 C floor, fit from closed-loop DFN residuals) and monitor it without claiming a guarantee.

C. Aging without an oracle, and why a bound is enough

Internal resistance grows with age [25], and a filter anchored to fresh dynamics under-predicts heat and voltage and turns unsafe. We estimate the resistance scale $\hat{s}_R$ online by inverting the measured overpotential [21], [22] and re-anchor the certificate every step, offsetting the thermal margin by the estimate. The estimator is *fail-safe*: it initializes worst-case and relaxes toward the measurement, staying conservative from step one on the same noisy sensors the policy reads. Its job is far weaker than identification, since resistance enters heat and voltage monotonically and a larger scale only makes the filter more cautious.

Proposition 2 (Monotone safety in the aging channels)**.** *Let aging act through two scalar channels. A resistance scale s_R multiplies the ohmic heat and voltage rise, and a cooling-loss factor $\gamma \ge 1$ divides the effective heat-transfer coefficient, so the temperature rise scales with γ. For the certified set $\mathcal{S} = \{T \le T_{\max}, V \le V_{\max}\}$, each one-step peak $g_c(s_R, \gamma, I)$, $c \in \{T, V\}$, is non-decreasing in the current I and in* each *channel s_R, γ, so the certified cap $I^\star(s_R, \gamma)$ is non-increasing in each. Hence for a plant with true channels $(s_R^\star, \gamma^\star)$, any $(\hat{s}_R, \hat{\gamma}) \succeq (s_R^\star, \gamma^\star)$ commands a current the plant already admits. Oracle-free safety on $\mathcal{S}$ therefore needs only a conservative* bound on each channel, *never the exact values.*

Proof. The thermal map is $T'{=}T{+}\frac{\Delta t}{C_{\text{th}}}(Q_{\text{gen}}{-}\frac{hA}{\gamma}(T{-}T_{\text{amb}}))$ with generation $Q_{\text{gen}}{=}I^2 R_\Omega s_R{+}Q_{\text{act}}(I){+}Q_{\text{rev}}$, and the terminal voltage is $V{=}\text{ocv}{+}I R_\Omega s_R{+}\eta_{\text{act}}(I){+}V_1$, and three partial derivatives settle it. Both $\partial_I g_T$ and $\partial_I g_V$ are non-negative, since the ohmic and activation terms rise with current. Both $\partial_{s_R} g_T$ and $\partial_{s_R} g_V$ are non-negative, since the resistance scale multiplies ohmic heat and voltage. And $\partial_\gamma g_T \ge 0$ whenever $T \ge T_{\text{amb}}$, since worse cooling adds heat. The endothermic $Q_{\text{rev}}{=}I\,T\,\mathrm{d}U/\mathrm{d}T$ stays below 24% of the irreversible heat over the admissible band and never overturns $\partial_I g_T \ge 0$, which holds for every I above 0.08 C, an order of magnitude below the cap's 0.70 C floor, verified numerically over 10–80% SOC. Each g_c is thus non-decreasing in I and in each channel, so $I^\star$ is non-increasing in each, and any $(\hat{s}_R, \hat{\gamma}) \succeq (s_R^\star, \gamma^\star)$ commands a current the plant admits. □

Reducing aging to these two channels drops nothing that matters, because capacity fade, OCV shift, and entropic and diffusion terms need no margin, since the projection re-solves at the true state and the full-DFN ensemble carries them, leaving only resistance and cooling to erode the T/V headroom monotonically.

The fail-safe initialization supplies the bound, starting at $\hat{s}_R$=1.8, the rated end-of-life resistance from the datasheet aging envelope, a fixed design constant known before deployment, not a per-cell oracle, so $\hat{s}_R \geq s_R^\star$ holds from step one across the rated range and meets Prop. 2's precondition by construction. A cell aged *past* rated end of life ($s_R^\star > 1.8$) falls outside it, exactly where a BMS should retire the cell. A higher assumed scale only throttles harder and delivers less charge, so the estimator is *safety-neutral*. By Prop. 1 any policy the filter projects stays safe, and the estimate's error costs SOC, never a violation.

The margin realizes both channels of Prop. 2, δ_T=δ_{T0}+$K(\hat{s}_R$−1)+$f(T$−$T_{\mathrm{amb}})$. The throttle (K=15) converts the resistance bound into thermal headroom, and the reserve (f=0.25) holds back a fraction f of the temperature rise, certifying any cooling loss up to $\hat{\gamma}$=1+f, a 20% fault. The gains are sized to that 20%-loss edge, not fitted to a failure. Of the design constants, $T_{\max}$, $V_{\max}$ and $s_R^\star$ are manufacturer limits and the rated end-of-life envelope, f is derived from the cooling fault to be covered, and only K is chosen by simulation. Sweeping $K \in [5, 25]$ and $f \in [0.1, 0.4]$ leaves zero violations in all 3969 runs of the aged, cooling-faulted grid, a safe region, not a tuned point; K is then set by delivered charge, which it costs monotonically, rather than by safety. The reserve also proves conservative, holding safety to a 33% cooling loss where it certifies 20%. We freeze the gains and test on a *disjoint* ensemble they never saw, and the filter stays safe on every held-out draw, including the joint noise/aging/out-of-model case a real BMS faces (Sec. VI, Table III).

The margin's price is aged charge. With the estimator in the loop, delivered SOC falls from 0.772 fresh to 0.589 at SOH 0.8 at zero violations; a non-adaptive filter keeps delivering 0.781 but violates T/V. Closing the loop against the DFN, the SOH 0.8 cell reaches 0.563 SOC within 3 points of the in-model value, ruling out a ROM artifact. The conservative bound alone gives a 0.533 floor, plus the estimator's ∼3 %. At rated end-of-life the filter trades roughly a quarter of fresh SOC for the margin.

D. Policy as a control, not a claim

Because the filter carries safety for *any* input, the policy is free. Its deployable default is a greedy governor requesting $I_{\max}$ every step. A PID-Lagrangian PPO actor–critic [29], [9] trained with the projection active is indistinguishable from it (Sec. VI). We therefore treat the governor as the method and PPO as an ablation.

V. Experimental Setup

The filter runs on a lumped ROM with three coupled parts [30]. A single-RC branch carries a temperature-dependent ohmic resistance, a lumped thermal state absorbs the ohmic, reaction, and reversible heat [31], and a surrogate tracks the plating overpotential [23]. Its parameters are fit to the Doyle–Fuller–Newman model [32], [33] under the parameterization of O'Kane et al. [27] by simulated charging sweeps with a held-out split (voltage RMSE 20.3 mV, held-out 21.3 mV, temperature RMSE 1.32 °C). Every safety claim is then validated out of sample by replaying the *realized* current in the full DFN in PyBaMM [34], so the ROM never self-reports safety, and worst-case out-of-distribution excursions are disclosed below.

TABLE I
Warm comparison at 25 C over 40 frozen CRN conditions.

Method	η_{sys}[%]	SOC	T_{pk}[°C]	Viol.[%]
CC–CV 1C	87.8	0.773	38.7	22
CC–CV 2C	84.4	0.803	53.5	100
MSCC	85.0	0.802	50.5	100
Offline opt.	85.9	0.798	44.5	0
PPO-Lag (no filter)	87.3	0.734	40.1	9
Ours	86.5	0.773	40.7	≤0.60

We compare CC–CV at several rates, multi-stage CC (MSCC), an offline time-optimal profile, a filter-free PID-Lagrangian PPO, and our filtered policy, all learned methods sharing the MDP, network, and budget. Metrics are system efficiency η_{sys} (cell energy over wall energy, including converter loss), delivered SOC at the deadline, DFN peak temperature, and the any-constraint violation rate over n episodes, reported with a Clopper–Pearson 95% upper bound [35]. The main aging comparison counts n=30 distinct scenarios, an honest rule-of-three bound of 10%, while the held-out ensembles use larger n.

VI. Results

A. Main comparison

Table I gives the warm comparison; the peak is DFN-replayed, and the violation column pairs each baseline's observed any-constraint rate with a T/V-only Clopper–Pearson bound for ours. Fast CC–CV and MSCC hit the target SOC but overshoot 45 °C in every episode. Conservative CC–CV 1C stays thermally cool (peak 38.7 °C) yet still *plates* in 22% of episodes (all its violations are plating, none thermal), a reminder that even a moderate sustained rate drives the anode negative at high SOC. Our filtered policy holds zero observed thermal and voltage violations (Clopper–Pearson upper bound ≤0.60%). It matches CC–CV 1C's SOC while removing that protocol's 22% plating (Fig. 3). A *predictive* controller banks more with foresight, the offline optimum reaching 0.798 by seeing the future and a causal MPC banking more at the cost of an online solver (Sec. VI-E). The filter also keeps headroom, sitting near 40.7 °C, the 45 °C limit *less* its cooling-robustness margin, so a cooling fault keeps slack; the offline optimum reserves none and rides to 44.5 °C. The filter-free PID-Lagrangian PPO [9] delivers less charge (0.734) and still violates 9%, the soft-penalty hedge the hard filter avoids.

The binding constraint flips with temperature. Warm, temperature binds and plating is slack. Cold (10 °C), the anode overpotential binds first and the plating cap tapers to 1.0 C, and the same filter handles both without retuning since the projection reads whichever constraint is closest. Cold is the harder regime, where our filter delivers 0.707 SOC at zero violations while both CC–CV rates violate and the causal gap widens from 3.0 % warm to 9.1 % cold.

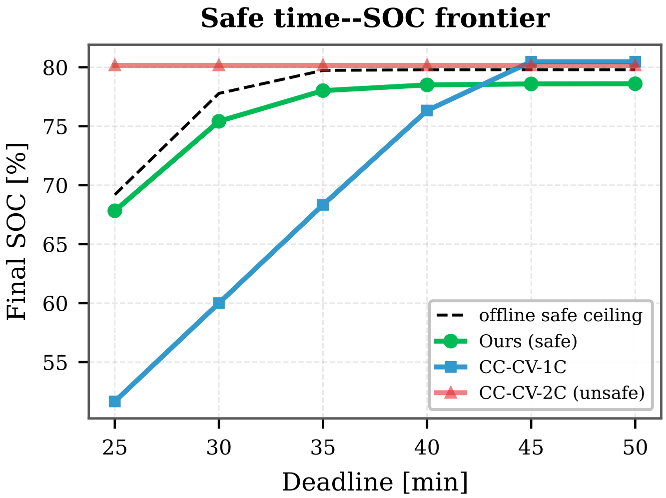

Fig. 3. Safe time–SOC frontier. Delivered SOC against the charge deadline at zero observed thermal or voltage violations. Our filter sits on the safe frontier, trailing only the non-causal offline optimum that reserves no cooling margin.

TABLE II
CAUSAL OPTIMALITY, DFN-REPLAYED (WARM, FRESH).

Method	SOC	T_{pk}[°C]	Anode[mV]	DFN-safe
Offline opt. (non-causal)	0.798	42.0	2.4	✓
Ref. governor (greedy)	0.768	40.0	14.6	✓
Ours (PPO)	0.768	40.0	14.5	✓

Table II shows the payoff of a policy-agnostic guarantee. The greedy governor and the trained PPO policy deliver the same SOC, peak, and violation rate, because the filter, not the policy, sets the operating point, so the simplest policy suffices and no learning is needed for safety. The governor is not globally optimal; the non-causal offline optimum banks 3.0 % more SOC by easing early with foresight. So across warm and cold the filter is the charge-maximal deployable choice at zero observed violations, and only an online solver or non-causal foresight does better.

B. Safety under aging, without a health oracle

On cells aged to SOH $\{1.0, \dots, 0.8\}$, the filter with the true aged model holds violations at 0%, while without adaptation it rises to 100.0% at SOH 0.8 and a filter-free PPO-Lagrangian to 83.3% (Fig. 4), the non-adaptive filter eroding its own thermal certificate as the cell ages. Closing the loop without an oracle, our online estimator holds violations at 0.0% across the SOH range, 0.0% under sensor noise, where the non-adaptive filter fails 100%. The aged-thermal margin is empirical out of model, so we test it statistically, and on a disjoint held-out ensemble it is safe on 156/156 draws (worst 41.9 °C, bound 1.9%) where the non-adaptive filter overheats. Table III collects every T/V stress test with its bound, including the joint case (noise, aging, and out-of-model DFN, bound 3.8% at n=78) and the cross-chemistry recalibration; each row counts violations over independent aged-DFN draws with the gains frozen, not i.i.d. cells. Safety thus holds across the rated aging range with no per-cell identification, the deployment property fresh-cell validation can never establish.

TABLE III
ALL T/V SAFETY EVIDENCE BY STRESS TEST, WITH SAMPLE SIZE, VIOLATIONS, AND A ONE-SIDED CLOPPER–PEARSON 95% UPPER BOUND.

Stress test	n	viol.	CP 95% [%]
Fresh cell, T/V (warm)	500	0	$\leq$0.6
Aged T/V, disjoint held-out	156	0	$\leq$1.9
Aged T/V, pooled	196	0	$\leq$1.5
Joint noise+aging+DFN	78	0	$\leq$3.8
Cross-param. aged T/V (Chen *et al.*)	60	0	$\leq$4.9
2nd chemistry, recalibrated (Ai *et al.*)	100	0	$\leq$3.0

C. Safety needs a bound, not identification

Fig. 5 closes the entire loop against the DFN and sweeps the resistance scale fed to the filter. The closed-loop *peak* temperature inherits Prop. 2's monotonicity, falling from 45.1 °C at $\hat{s}_R$=1.0 to 26.9 °C at $\hat{s}_R$=2.2 (SOH 0.8), and the safe boundary sits near $\hat{s}_R\approx$1.1, far below the true $s^\star_R$=1.8. The fail-safe init thus clears it with room to spare, so oracle-free safety rests on the bound, not on the DFN matching the ROM, and the resistance estimator sets only charge speed. The one orthogonal fault, an unmodeled cooling loss, is handled by the margin's second channel.

The estimate *under*-shoots by end of charge ($\hat{s}_R\to$1.44 vs $s^\star_R$=1.80), which would be unsafe were the final estimate the certificate. It is not, because the fail-safe init pins $\hat{s}_R$=1.8 $\geq s^\star_R$ through the high-current phase and the dip below $s^\star_R$ comes only at 29 min, once current has fallen to 0.7 C with the cell 13.8 °Cbelow the limit. The theorem carries the stressful phase, the margin the tail, and the estimate's error costs charge, never safety.

D. High-fidelity DFN validation

On the plating channel, our nominal warm and cold profiles hold minimum anode potentials of +13.2 mV and +10.8 mV, both positive, where CC–CV 2C drives the cold anode to −62.2 mV. Under an adversarial out-of-distribution ensemble that breaks the plating calibration, the cap holds on 99/100 warm and 97/100 cold draws (worst -2.0 mV, cold bound 7.6%), the few breaches landing in the cold corner the monotonicity flags.

The certified margin is not an artifact of the reference degradation set [27]. Replaying our aged profiles across a perturbed Chen et al. ensemble (same 4.2 V/5 Ah spec) holds T/V on all 60/60 draws (bound 4.9%); recalibrating the *entire* ROM (OCV, overpotential, thermal) to the Enertech cell of Ai et al. [36] (2.28 Ah, a different manufacturer, voltage RMSE 26.5 mV) keeps the closed-loop filter safe on all 100/100 aged draws (bound 3.0%, Table III). That cell runs cool (peak 28 °C), so voltage binds and the test exercises the *voltage* certificate's transfer. Plating recalibration would need kinetics that set lacks, so it stays LG M50, and hardware remains out of scope. The certificate therefore survives high-fidelity electrochemistry and a change of chemistry, evidence that safety follows from the method and not a tuned parameter set.

E. Solver-free feasibility versus MPC

The projection is the only online computation carrying safety, a bisection of a few ROM forward maps with no matrix

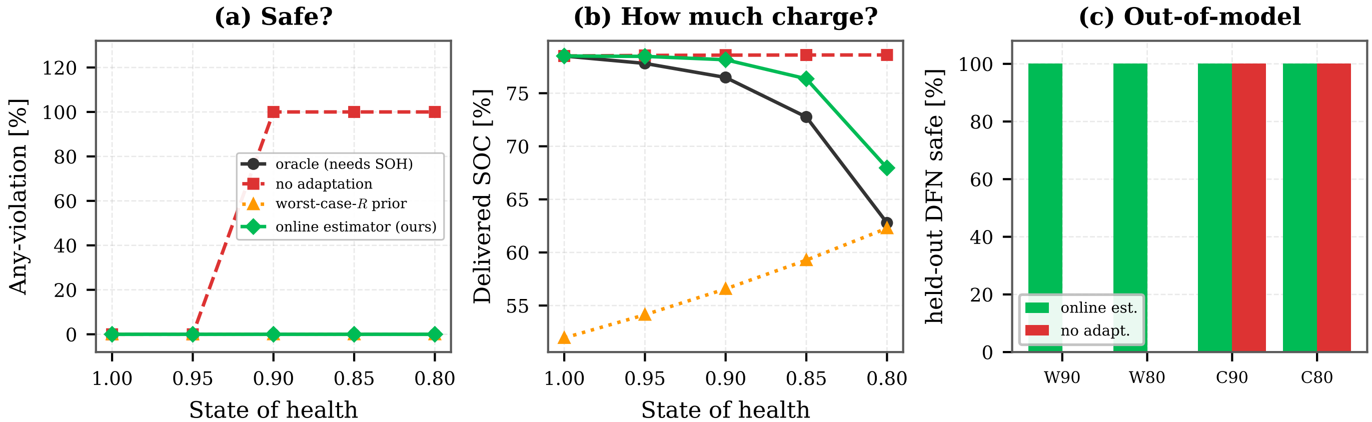

Fig. 4. Deployment across the SOH range. **(a)** among the filtered variants, only "no adaptation" violates, and early (55% of episodes at SOH 0.9, 100% at 0.8). Oracle, worst-case-R prior, and our estimator stay at 0% in model. **(b)** the worst-case prior wastes charge on healthy cells while the estimator recovers the oracle's charge. **(c)** held-out aged-DFN ensemble, where the estimator stays safe out of model.

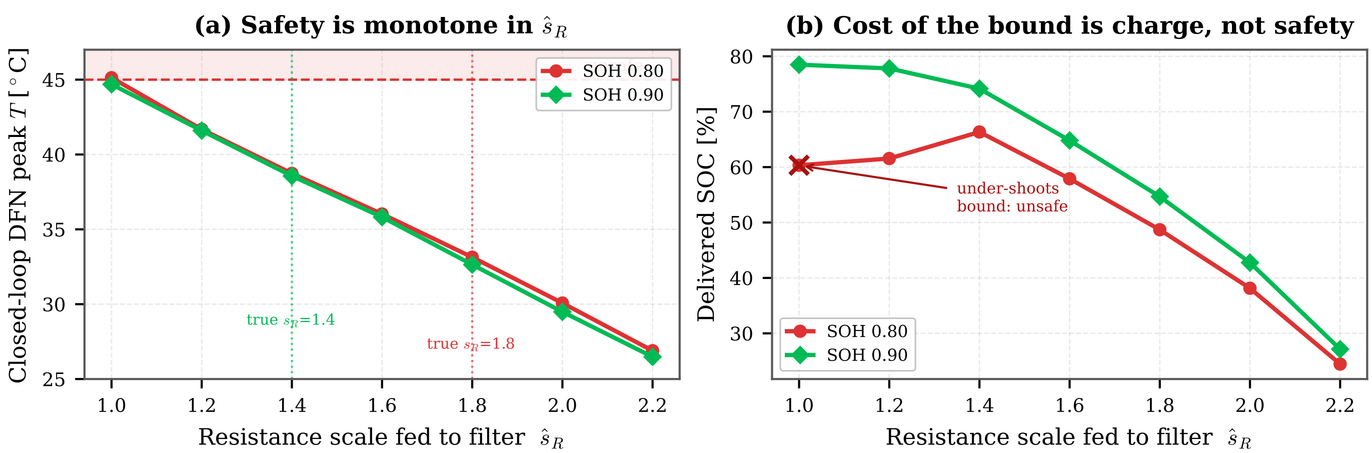

Fig. 5. Closed-loop against the DFN, safety is monotone in the resistance scale the filter is told to use (Prop. 2). **(a)** peak temperature falls as $\hat{s}_R$ grows. Every value past a boundary near 1.1, well below the true aged $s_R^\star$ (dotted), is safe, so a conservative bound certifies safety without identifying $s_R^\star$. **(b)** the bound's only price is delivered charge, which is why an online estimate still earns its keep, for speed. Under-shooting the bound ($\hat{s}_R$=1.0, $\times$) is the lone unsafe point.

factorization. It costs $O(\log(1/\epsilon))$ model evaluations for current resolution ϵ, and because the iteration count is fixed rather than convergence-tested, that bound is met on every input: at most 20 closed-form evaluations, $\sim$800 flops, and 904 B of state and constants with no dynamic allocation, 0.9 ms worst-case in the reference implementation and orders of magnitude inside the 30 s step. The tolerance after k halvings is $I_{\max}/2^k$, 38 μA at k=18; since the bisection maintains lo $\leq I^\star$, a coarser tolerance only under-commands, so safety is independent of k and only delivered charge varies, saturating by $k\approx$12. Its safety logic is 36 numpy-only source lines a functional-safety audit can read in full, where an embedded QP instead places a compiled ADMM solver (39 bundled C source and header files) in the trusted computing base. Against a receding-horizon MPC ladder, nominal MPC (fresh R) violates 100% under aging, while an MPC given the *same* fail-safe bound is oracle-free and safe (0%), as is an oracle MPC (0%). Oracle-freedom is thus Prop. 2's doing, shared by any bounded controller, not the filter's. Because it is a bound and not an estimate, it covers what it bounds and degrades gracefully past it: raising the true resistance scale beyond the rated $s_R^\star$ leaves every in-envelope cell safe with margin in hand, and the first violation appears only at s_R=3.0, 1.7$\times$ the rated bound and far beyond the point a BMS retires the cell. Unique to the filter is Prop. 1, replacing the solver with a fixed 18-iteration bisection of the *exact* constraint, ceding some charge to the foresighted MPC (0.724 vs 0.779) for that simplicity. The trade is deliberate, buying MPC-class safety without a solver in the trusted computing base for a few points of charge.

VII. Conclusion

We built a real-time safety filter for fast charging that needs no online solver and no state-of-health oracle, and two facts carry it. Zero current is a universal safe backup for temperature and voltage, collapsing the predictive safety filter to a scalar bisection. Safety is monotone in the two aging channels, resistance growth and cooling loss, so a conservative datasheet bound certifies it and no parameter is identified. The bound carries safety, and the online estimator only buys charge speed.

The filter records zero observed temperature and voltage violations over hundreds of DFN-confirmed episodes where fast

CC–CV overshoots 45 °C every time. It stays safe on 156/156 disjoint aged draws (Clopper–Pearson bound 1.9%) and under a full recalibration of the ROM to a second cell chemistry, so the guarantee is a property of the method and not one cell. It delivers the charge of CC–CV at 1C while removing that protocol's plating, and a trivial greedy governor matches a trained RL policy on every metric, which isolates the filter as the safety carrier. The certified path is 36 numpy-only source lines, small enough to audit.

The certificate is ROM-relative on volume-averaged temperature, the evidence is simulation over an aged-DFN distribution and not a hardware rating, and plating is enforced and monitored, not certified. Hardware validation, a predictive plating constraint, an age-aware cap, and a pack-level extension are the natural next steps, in that order: hardware validation is what converts a ROM-relative certificate into a rated one, and a predictive plating constraint is what would move plating from enforced to certified. We release the filter, ROM, executable proposition checks, and a one-command reproduction as an MIT-licensed reference implementation at https://github.com/keshavkrishnan08/safe-charge.

REFERENCES

[1] A. Tomaszewska, Z. Chu, X. Feng *et al.*, "Lithium-ion battery fast charging: A review," *eTransportation*, vol. 1, p. 100011, 2019.

[2] T. Waldmann, M. Wilka, M. Kasper, M. Fleischhammer, and M. Wohlfahrt-Mehrens, "Temperature dependent ageing mechanisms in lithium-ion batteries," *J. Power Sources*, vol. 262, pp. 129–135, 2014.

[3] F. Wang *et al.*, "Deep reinforcement learning based fast charging of lithium-ion batteries with different penalty coefficients," in *IECON 2022 - 48th Annual Conf. IEEE Industrial Electronics Society*, 2022.

[4] S. Chowdhury, S. Al-Wahaibi, and Q. Lu, "Adaptive safe reinforcement learning-enabled optimization of battery fast-charging protocols," *AIChE Journal*, 2025, arXiv:2406.12309.

[5] P. M. Attia, A. Grover, N. Jin *et al.*, "Closed-loop optimization of fast-charging protocols for batteries with machine learning," *Nature*, vol. 578, pp. 397–402, 2020.

[6] K. P. Wabersich and M. N. Zeilinger, "A predictive safety filter for nonlinear learning-based control," *Automatica*, vol. 129, p. 109597, 2021.

[7] D. Angeli and E. D. Sontag, "Monotone control systems," *IEEE Transactions on Automatic Control*, vol. 48, no. 10, pp. 1684–1698, 2003.

[8] S. Coogan and M. Arcak, "Efficient finite abstraction of mixed monotone systems," in *18th ACM Int. Conf. on Hybrid Systems: Computation and Control (HSCC)*, 2015, pp. 58–67.

[9] A. Stooke, J. Achiam, and P. Abbeel, "Responsive safety in reinforcement learning by PID lagrangian methods," in *ICML*, 2020.

[10] J. Achiam, D. Held, A. Tamar, and P. Abbeel, "Constrained policy optimization," in *ICML*, 2017.

[11] M. Alshiekh, R. Bloem, R. Ehlers, B. Könighofer, S. Niekum, and U. Topcu, "Safe reinforcement learning via shielding," in *AAAI*, 2018.

[12] G. Dalal, K. Dvijotham, M. Vecerik, T. Hester, C. Paduraru, and Y. Tassa, "Safe exploration in continuous action spaces," in *arXiv:1801.08757*, 2018.

[13] Z. Wei, Z. Quan, J. Wu, Y. Li, J. Pou, and H. Zhong, "Deep deterministic policy gradient-DRL enabled multiphysics-constrained fast charging of lithium-ion battery," *IEEE Trans. Industrial Electronics*, vol. 69, no. 3, pp. 2588–2598, 2022.

[14] K. Kim *et al.*, "Reinforcement learning for battery charging control," *IEEE Trans. Industrial Electronics*, vol. 67, no. 8, pp. 6825–6835, 2020.

[15] F. Blanchini and S. Miani, *Set-Theoretic Methods in Control*. Birkhäuser, 2008.

[16] A. D. Ames, S. Coogan, M. Egerstedt, G. Notomista, K. Sreenath, and P. Tabuada, "Control barrier functions: Theory and applications," *European Control Conference (ECC)*, 2019.

[17] E. Gilbert and I. Kolmanovsky, "Nonlinear tracking control in the presence of state and control constraints: A generalized reference governor," *Automatica*, vol. 38, no. 12, pp. 2063–2073, 2002.

[18] S. Bansal, M. Chen, S. Herbert, and C. J. Tomlin, "Hamilton–jacobi reachability: A brief overview and recent advances," in *IEEE Conf. Decision and Control (CDC)*, 2017.

[19] M. D. Berliner *et al.*, "Real-time optimal fast charging of li-ion batteries with varying temperature and charging behaviour constraints," *J. Energy Storage*, 2021.

[20] S. J. Moura, F. B. Argomedo, R. Klein, A. Mirtabatabaei, and M. Krstic, "Battery state estimation for a single particle model with electrolyte dynamics," *IEEE Trans. Control Syst. Technol.*, vol. 25, no. 2, pp. 453–468, 2017.

[21] G. L. Plett, *Battery Management Systems, Volume 1: Battery Modeling*. Artech House, 2015.

[22] W. Waag, C. Fleischer, and D. U. Sauer, "Critical review of the methods for monitoring of lithium-ion batteries in electric and hybrid vehicles," *J. Power Sources*, vol. 258, pp. 321–339, 2014.

[23] X.-G. Yang, Y. Leng, G. Zhang, S. Ge, and C.-Y. Wang, "Modeling of lithium plating induced aging of lithium-ion batteries," *J. Power Sources*, vol. 360, pp. 28–40, 2017.

[24] P. Arora, R. E. White, and M. Doyle, "Capacity fade mechanisms and side reactions in lithium-ion batteries," *J. Electrochem. Soc.*, vol. 145, no. 10, pp. 3647–3667, 1998.

[25] N. Kirkaldy, M. A. Samieian, G. J. Offer, M. Marinescu, and Y. Patel, "Lithium-ion battery degradation: Comprehensive cycle ageing data and analysis for commercial 21700 cells," *J. Power Sources*, vol. 603, p. 234185, 2024.

[26] C.-H. Chen, F. B. Planella, K. O'Regan, D. Gastol, W. D. Widanage, and E. Kendrick, "Development of experimental techniques for parameterization of multi-scale lithium-ion battery models," *J. Electrochemical Society*, vol. 167, no. 8, p. 080534, 2020.

[27] S. E. J. O'Kane, W. Ai, G. Madabattula, D. Alonso-Alvarez, R. Timms, V. Sulzer, J. S. Edge, B. Wu, G. J. Offer, and M. Marinescu, "Lithium-ion battery degradation: How to model it," *Physical Chemistry Chemical Physics*, vol. 24, pp. 7909–7922, 2022.

[28] E. Altman, *Constrained Markov Decision Processes*. CRC Press, 1999.

[29] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," in *arXiv:1707.06347*, 2017.

[30] J. Newman and K. E. Thomas-Alyea, *Electrochemical Systems*, 3rd ed. Wiley, 2004.

[31] D. Bernardi, E. Pawlikowski, and J. Newman, "A general energy balance for battery systems," *J. Electrochem. Soc.*, vol. 132, no. 1, pp. 5–12, 1985.

[32] M. Doyle, T. F. Fuller, and J. Newman, "Modeling of galvanostatic charge and discharge of the lithium/polymer/insertion cell," *J. Electrochem. Soc.*, vol. 140, no. 6, pp. 1526–1533, 1993.

[33] T. F. Fuller, M. Doyle, and J. Newman, "Simulation and optimization of the dual lithium ion insertion cell," *J. Electrochem. Soc.*, vol. 141, no. 1, pp. 1–10, 1994.

[34] V. Sulzer, S. G. Marquis, R. Timms, M. Robinson, and S. J. Chapman, "Python battery mathematical modelling (PyBaMM)," *J. Open Research Software*, vol. 9, no. 1, 2021.

[35] C. J. Clopper and E. S. Pearson, "The use of confidence or fiducial limits illustrated in the case of the binomial," *Biometrika*, vol. 26, no. 4, pp. 404–413, 1934.

[36] W. Ai, L. Kraft, J. Sturm, A. Jossen, and B. Wu, "Electrochemical thermal-mechanical modelling of stress inhomogeneity in lithium-ion pouch cells," *J. Electrochem. Soc.*, vol. 167, no. 1, p. 013512, 2020.